Keshav Sharma

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EDUCATION

Vellore Institute of Technology **Chennai, Tamil Nadu**

Bachelor of Technology in Computer Science (Data Science Specialisation) *Graduation: May 2027*

SKILL SUMMARY

Languages: Java, C/C++, Python, SQL, JavaScript, R
Software Engineering: Data Structures & Algorithms, Object-Oriented Programming (OOP), Operating Systems, Database Management Systems (DBMS), Computer Networks, REST APIs, Multithreading, Concurrency, Design Patterns
Backend & Frameworks: Flask, FastAPI, Node.js, React.js
Databases & Storage: MySQL, SQLite, Redis
Tools & Platforms: Git, GitHub, GitHub Actions, Docker, VS Code, IntelliJ IDEA, Jupyter Notebook

PROJECT EXPERIENCE

***RedisLite* |LINK**

- Engineered RedisLite, a Redis-inspired in-memory key-value database in Python featuring a custom RESP protocol parser, multithreaded TCP server, thread-safe storage engine, and an O(1) custom LRU cache built using a hash map and doubly linked list, with lazy and active TTL expiration.
- Designed a four-layer architecture (TCP transport, storage engine, cache/TTL, persistence) implementing Write-Ahead Logging (WAL), atomic log compaction using os.replace(), and automatic crash recovery; validated the system with 46 automated tests and benchmarked 619K SET ops/sec for in-memory operations and 2.8K ops/sec with durable WAL persistence.

***API Rate Limiter Library* |LINK**

- Implemented a production-ready API rate-limiting library in Python featuring five industry-standard throttling algorithms, Redis integration, and framework middleware for Flask and FastAPI applications.
- Designed a modular architecture with storage abstraction, concurrent request handling, automated testing (30 test cases), and package distribution through PyPI for public installation and reuse.

***AI-Powered Expense Receipt Scanner & Analytics Platform* |LINK**

- Designed and implemented a comprehensive full-stack expense management platform that automates receipt processing; enhanced processing time by 40% while ensuring accuracy through EasyOCR, OpenCV, and advanced NLP-based parsing techniques.
- Implemented and deployed a scalable application using FastAPI, React, Vite, SQLAlchemy, SQLite, JWT Authentication, and Google OAuth, featuring expense analytics, budget tracking, category-wise spending insights, receipt management (CRUD), Excel export, and a responsive dashboard hosted on Render and Vercel.

***Real-Time Stock Anomaly Detector* | LINK**

- Built an end-to-end anomaly detection pipeline on 2 years of NSE stock data, engineering 9 financial features (RSI, Bollinger Bands, volume ratio); deployed dual models — Isolation Forest (2.01% anomaly rate) and LSTM Autoencoder — achieving temporal pattern detection across 448 trading sessions with an interactive Streamlit dashboard
- Integrated FinBERT financial sentiment analysis on live news headlines to correlate price anomalies with news tone, identifying high-confidence anomaly events such as the Apr 2025 market selloff (score −0.6332) and Jan 2026 peak reversal through cross-validation against NSE announcements

Certifications

Oracle Cloud Infrastructure 2025 Certified Generative AI Professional | |LINK ***2026***

- Demonstrated proficiency in Generative AI, Large Language Models (LLMs), Prompt Engineering, Retrieval-Augmented Generation (RAG), AI Agents, responsible AI practices, and building AI applications using Oracle Cloud Infrastructure (OCI).

Oracle Cloud Infrastructure 2025 Certified AI Foundations Associate |LINK ***2026***

- Demonstrated foundational knowledge of AI, Machine Learning, Generative AI, Large Language Models, and Oracle Cloud AI services.

VOIS – Vodafone Idea Foundation (virtual internship)| LINK ***2025***

- Completed an AICTE-certified virtual internship focused on LLM-based data analysis using Airbnb and Netflix datasets.

BCG Data Science Job Simulation |LINK ***2025***

- Performed customer churn analysis using Python (Pandas, NumPy), built and optimised a Random Forest model achieving 50% recall, and delivered data-driven business recommendations through visualisations and executive reporting.